P525/2
CHEMISTRY
Paper 2
(Practical)
July/Aug. 2026
2 Hours

AITEL END OF TERM II ASSESSMENTS 2026

Uganda Advanced Certificate of Education

CHEMISTRY

SENIOR FIVE

PAPER 2
(Practical)
2 Hours

INSTRUCTIONS TO CANDIDATES:

*This examination paper consists of one **compulsory** examination item.*

*Responses to these items **must** be written in the answer booklet provided.*

Begin each item on a fresh page.

All working must be clearly shown. Graph papers will be provided.

Mathematical tables and silent non-programmable scientific calculators may be used.

*You are **not** allowed to use reference books (i.e, text books, booklets on qualitative and quantitative analysis, etc.)*

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ITEM 1

A local construction research institute near an industrial zone is exploring the use of a newly processed inorganic solid, **Solid Y**, as an alternative binder for building materials. The enthalpy of formation of Solid Y is a key measure of its thermal stability and durability in the environment, but it cannot be determined directly by simple calorimetry. This, however, can be determined indirectly using **Hess's Law of Constant Heat Summation**.

Senior Five learners from a neighboring secondary school visited the research facility to assist in evaluating the stability of Solid Y before it can be recommended for construction, but they do not know how this can be done.

The laboratory technician provided the learners with two solid samples, Solid X and Solid Y, and an aqueous reagent, FA1. The technician advised them to react approximately **1.2 g** of Solid X and **2.0 g** of Solid Y separately with not less than **60 cm^3** of FA1, measuring the temperature of the mixture at specific time intervals. However, even with this advice, the learners could not determine the stability of Solid Y and therefore need your help.

Both Solid X and Solid Y react separately with FA1 according to the following thermochemical relationships:

Solid X(s) + 2FA1(aq) ⟶ Aqueous Salt + Gas Z(g) (ΔH_1)

Solid Y(s) + 2FA1(aq) ⟶ Aqueous Salt + H_2O(l) (ΔH_2)

Standard enthalpy of formation of liquid water:

Gas Z(g)+12O_2(g)⟶H_2O(l) ΔH_3= −285.8 kJ mol−1

Additional Information

- Specific heat capacity of the reaction mixture = **4.2 J g^{-1} °C^{-1}**
- Density of the reaction mixture = **1.0 g cm^{-3}**
- Molar mass of Solid X = **24.0 g mol^{-1}**
- Molar mass of Solid Y = **40.0 g mol^{-1}**

You are provided with: Solid X, Solid Y and FA1

Task:

Design and carry out an investigation, then use your findings to help the Senior Five learners. *(Your response should be in form of a scientific report. A minimum of six readings are required).*

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Great care should be taken to ensure that the information given below does not reach the candidates either directly or indirectly.

a) 2 insulated plastic beakers)
b) 1 Thermometer
c) 1 Measuring cylinder (50 cm^3 or 100 cm^3)
d) 1 Stop clock / stop watch
e) 1 Stirring glass rod
f) 100 cm^3 of distilled water
g) Easy access to a weighing balance reading to at least 1 decimal places
h) **Solid X (1.2 g):** Clean Magnesium metal powder.
i) **Solid Y (2.0 g):** Dry Magnesium oxide powder.
j) **FA1 (200 cm^3):** 2.0 M Hydrochloric acid solution (HCl). Prepared by diluting 166 cm^3 of concentrated hydrochloric acid (12 M) with distilled water to make 1.0 litre of solution.

END

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